

```

`timescale 1ns / 1ps
////////////////////////////////////////////////////////////////
module MUX(I0,I1,I2,I3,s,out);

input I0,I1,I2,I3;
input [1:0]s;
output out;

wire [1:0]sn;
wire [3:0]A0;

not n0 (sn[0],s[0]);
not n1 (sn[1],s[1]);

and a0 (A0[0],I0,sn[0],sn[1]);
and a1 (A0[1],I1,s[0],sn[1]);
and a2 (A0[2],I2,sn[0],s[1]);
and a3 (A0[3],I3,s[0],s[1]);

or o (out,A0[0],A0[1],A0[2],A0[3]);

endmodule

```

```

`timescale 1ns / 1ps

////////////////////////////////////////////////////////////////

module Test;

// Inputs
reg I0;
reg I1;
reg I2;
reg I3;
reg [1:0] s;

// Outputs
wire out;

```

```

// Instantiate the Unit Under Test (UUT)
MUX uut (
    .I0(I0),
    .I1(I1),
    .I2(I2),
    .I3(I3),
    .s(s),
    .out(out)
);

initial begin
    // Initialize Inputs
    I0 = 0;
    I1 = 0;
    I2 = 0;
    I3 = 0;
    #100;
    // Wait 100 ns for global reset to finish

    end
    always begin
        #10;
        I0 = ~I0;
        #2;
    end

    always begin
        #10;
        I1 = ~I1;
        #4;
    end
    end
    always begin
        #10;
        I2 = ~I2;
        #10;
    end
    end
    always begin
        #10;
        I3 = ~I3;
        #15;
    end
    end

    initial begin

```

```
s = 2'b00;  
#100;  
s = 2'b01;  
#100;  
s = 2'b10;  
#100;  
s = 2'b11;  
#100;  
end
```

```
endmodule
```